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Molecular Simulation Study of the Performance of Supported Ionic Liquid Phase Materials for the Separation of Carbon Dioxide from Methane and Hydrogen

Samir Budhathoki,[†] Jindal K. Shah,[‡] and Edward J. Maginn^{*,†}

*Department of Chemical and Biomolecular Engineering, Notre Dame, Indiana 46556,
United States, and School of Chemical Engineering, Stillwater, Oklahoma 74078, United
States*

E-mail: ed@nd.edu

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*To whom correspondence should be addressed

[†]University of Notre Dame

[‡]Oklahoma State University

Abstract

Molecular dynamics and Gibbs ensemble Monte Carlo simulations were used to compute the self-diffusion coefficients and solubilities of CO₂, CH₄ and H₂ in model membranes consisting of slit pores with diameters of 2 nm and 5 nm. Solubility selectivities, diffusion selectivities and permselectivities of CO₂ for binary gas mixtures of CO₂/CH₄ and CO₂/H₂ were also computed. The calculations were repeated for the same pores filled with the ionic liquid (IL) 1-n-butyl-3-methylimidazolium bis(trifluoromethyl sulfonyl)imide ([C₄mim]⁺[Tf₂N]⁻) and for bulk IL. The bulk IL system was used as a model for a supported ionic liquid membrane separator having large pores, while the confined IL systems were used to assess whether extreme nanoconfinement of ILs has an effect on permselectivity. Permselectivities were about a factor of ten higher in all the IL systems compared to the empty nanopores. Nanoconfinement tends to increase the solubility and decrease the diffusivity of all gases relative to the bulk IL. The bulk IL has significantly higher solubility selectivity for CO₂ over CH₄ and H₂ relative to the empty pores, and nanoconfinement of the IL further increases solubility selectivity a modest amount. Diffusion selectivities in the nanoconfined IL for CO₂ over CH₄ are slightly enhanced relative to bulk IL, but are slightly smaller for CO₂ over H₂. The net result is that nanoconfinement of the IL is predicted to slightly increase permselectivity for CO₂ over CH₄ but have little effect on the permselectivity of CO₂ over H₂ when compared to bulk IL. Although the IL leads to significantly enhanced permselectivities of CO₂ compared to the empty nanopores, gas diffusivities are more than two orders of magnitude smaller in the IL when compared to the empty nanopores. This suggests that while the use of this IL in a supported ionic liquid membrane separator will lead to enhanced selectivities, the overall permeation rate may be reduced relative to a conventional membrane if diffusion in the pores is rate limiting.

Introduction

Ionic liquids (ILs) are salts with melting points below 100 °C. They possess several favorable properties that can be exploited for different applications.¹⁻⁶ In recent years, many studies have shown promising results regarding the use of ILs for separation of CO₂ from flue gas and natural gas.^{3,4,7-11} Unfortunately, many ILs have viscosities that are two or more orders of magnitude higher than traditional organic solvents. The high viscosity of ILs results in a significant increase in interfacial mass transfer resistance and an equally large reduction in the diffusion coefficient of the gas. This impedes the use of bulk IL as a gas separation agent in a conventional absorber-stripper configuration. In addition, many ILs are more expensive than traditional solvents, which suggests that strategies that minimize the amount of IL needed will be beneficial.

One way to overcome the mass-transfer limitations in highly viscous fluids and at the same time reduce the amount of material used is to employ thin films. Supported liquid membranes (SLMs) can be formed by contacting a membrane or porous support with a liquid that has a high affinity for one of the species to be separated in a mixture. One of the drawbacks of SLMs in gas separation processes is that the finite vapor pressure of the liquid results in the loss of liquid over time. Because ILs have extraordinarily low vapor pressures and can exhibit high solubilities for gases such as CO₂, supported ionic liquid membranes (SILMs) are now receiving much attention.¹²⁻¹⁷

Gas separation performance of membranes may be described using a solution-diffusion model in which the permeability of species i , P_i is given as the product of the solubility (S_i) and the diffusivity (D_i). Hence, the permselectivity for species i over species j , β_{ij}^P , is defined as

$$\beta_{ij}^P = \frac{P_i}{P_j} = \beta_{ij}^D \times \beta_{ij}^S \quad (1)$$

where β_{ij}^D and β_{ij}^S are diffusion selectivity and solubility selectivity at total pressure P and

temperature T , defined as

$$\beta_{ij}^D = \left(\frac{D_i}{D_j} \right)_{P,T} \quad (2)$$

$$\beta_{ij}^S = \left(\frac{S_i}{S_j} \right)_{P,T} = \left(\frac{x_i/y_i}{x_j/y_j} \right)_{P,T} \quad (3)$$

where x and y are mole fractions of species in the liquid and gas phase, respectively.

A number of studies have been performed with different classes of ILs to study the gas separation performance of SILMs. Scovazzo and co-workers have reported^{12,13} the superior CO₂/N₂ and CO₂/CH₄ permselectivities in SILMs compared to standard polymers. Similarly, an experimental study by Myers et al.¹⁴ on SILM with amine functionalized IL has shown that CO₂ selectivity over H₂ improved with increasing temperature unlike that in bulk IL. This work showed that there is an interplay between the solubility selectivity and diffusion selectivity of the gases. Cao et al.¹⁵ investigated CO₂ absorption in SILMs made up of different imidazolium-based ILs and observed excellent mass transfer efficiency and CO₂ absorption rate. Kasahara et al.¹⁸ employed ILs having amino acid anions in a facilitated transport membrane designed to separate CO₂ under dry conditions.

One of the main problems with SILMs in industrial processes is their inability to handle large pressure differentials. Weak capillary forces acting between the support and IL can be inadequate to contain the IL within the pores, leading to IL being displaced from the support when there are large pressure differentials across the membrane. SILMs have been shown to fail at a pressure difference as low as 2 bar across the membrane.¹⁹ Additionally, typical SILMs reported in the literature are quite thick, 150 μ m or more,^{4,19–21} and high-throughput industrial processes will require SILMs of less than 1 μ m thickness.^{22,23} Recently the concept of supported ionic liquid phase (SILP) materials for gas separation technologies has been investigated.^{24,25} In SILP materials, a thin film of IL is dispersed over the inner surface of a material with nanometer size pores.

The development of gas separation technologies such as those described above require

a more detailed understanding of gas absorption and diffusion in ILs. It is also critical to understand how extreme confinement, such as occurs in SILPs, might alter the properties of the IL and thus permeability. Several studies have already shown that physical, transport and sorption properties of ILs and solute gases change significantly upon extreme confinement. Chen et al.²⁶ and Dong et al.²⁷ have reported an increase in the melting temperature of the IL 1-n-butyl-3-methylimidazolium hexafluorophosphate ($[\text{C}_4\text{mim}]^+ [\text{PF}_6]^-$) upon confinement in carbon nanotubes. Neouze and Schauer,²⁸ Sha et al.,²⁹ and Bovio et al.³⁰ have reported that there are surface and confinement induced phase changes in different ILs. Shi and Sorescu³¹ observed an increase in the self-diffusion coefficients and solubilities of CO_2 and H_2 in 1-n-hexyl-3-methylimidazolium bis(trifluoromethyl sulfonyl)imide ($[\text{C}_6\text{mim}]^+ [\text{Tf}_2\text{N}]^-$) confined in carbon nanotubes. The self-diffusion coefficient of confined $[\text{C}_6\text{mim}]^+ [\text{Tf}_2\text{N}]^-$ was found to be 1-2 orders of magnitude greater than in the bulk liquid phase. In another study, Shi and Luebke³² have reported increased solubilities and self-diffusion coefficients of CO_2 , H_2 and N_2 in $[\text{C}_6\text{mim}]^+ [\text{Tf}_2\text{N}]^-$ confined in silica nanopores as opposed to bulk $[\text{C}_6\text{mim}]^+ [\text{Tf}_2\text{N}]^-$. The self-diffusion coefficient of confined $[\text{C}_6\text{mim}]^+ [\text{Tf}_2\text{N}]^-$ was reported to be higher than the corresponding bulk liquid diffusivity at 298 K, but lower than the bulk liquid diffusivity at 373 K. Similarly, Singh et al.³³ and Rajput et al.³⁴ reported heterogeneous dynamics of ILs when confined in a graphite slit pore. Li et al.³⁵ investigated the dynamic behavior of $[\text{C}_4\text{mim}]^+ [\text{Tf}_2\text{N}]^-$ in a silica mesopore and a carbon mesopore at different $[\text{C}_4\text{mim}]^+ [\text{Tf}_2\text{N}]^-$ pore loadings and temperatures. The dynamics of $[\text{C}_4\text{mim}]^+ [\text{Tf}_2\text{N}]^-$ in the silica mesopore changed significantly as a function of IL pore loading, while it displayed a weak temperature dependence. On the other hand, the dynamics of $[\text{C}_4\text{mim}]^+ [\text{Tf}_2\text{N}]^-$ in the carbon mesopore displayed a strong temperature dependence and a weak dependence on pore loading. Experimental investigations of different ILs confined in mesoporous silica gels have revealed that nanoconfined ILs have remarkably low specific heat capacities, disordered vibrational conformations, greater interactions with hydrocarbon solutes and enhanced fluorescence emission compared to the corresponding bulk ILs.³⁶ Iacob et al.³⁷ have experi-

mentally determined that the dynamics of 1-n-butyl-3-methylimidazolium tetrafluoroborate ($[\text{C}_4\text{mim}]^+ [\text{BF}_4]^-$) confined in silica nanopores is enhanced by around two orders of magnitude relative to the bulk IL. Close et al.³⁸ found that CO_2 permeability was enhanced in ILs supported on alumina nanopores compared to the permeability in bulk IL. Pulsed field gradient nuclear magnetic resonance investigations of diffusion in $[\text{C}_4\text{mim}]^+ [\text{Tf}_2\text{N}]^- \text{-CO}_2$ mixtures in the mesopores of KIT-6 silica revealed that the self-diffusion coefficients of the cation are reduced by a factor of two compared to that in bulk IL- CO_2 mixtures.³⁹

Based on these studies, it is clear that confinement can change the sorption and diffusion properties of ILs but there are no simple rules. A fundamental understanding on how confinement impacts the sorption and transport properties of ILs and various solutes is needed to advance SILP and SILM technology. The objective of this paper is to use molecular simulations to compute the solubilities and self-diffusion coefficients of pure and binary mixtures of CO_2 , CH_4 and H_2 in graphite slit pores having diameters of 2 nm and 5 nm (representative of nanoporous materials used in SILPs), in the bulk liquid phase of the IL $[\text{C}_4\text{mim}]^+ [\text{Tf}_2\text{N}]^-$ and in the IL confined in the slit pores. By invoking the solution-diffusion mechanism, permselectivities of CO_2 over CH_4 and H_2 can be determined for each system using eq 1. The bulk liquid permeabilities represent the values expected for IL in large pore membranes (i.e. SILMs) where confinement effects are minimal. The confined IL permeabilities (representative of the performance expected in SILPs) will be compared to the bulk liquid permeabilities to assess whether confinement enhances or reduces permselectivity. Finally, comparison will be made to the permselectivities of the empty slit pores to assess how much the IL affects permselectivity.

Simulation Details

Force Field

An all-atom force field with following functional form was used

$$\begin{aligned}
 V_{\text{tot}} = & \sum_{\text{bonds}} k_b (r - r_0)^2 + \sum_{\text{angles}} k_\theta (\theta - \theta_0)^2 \\
 & + \sum_{\text{dihedrals}} k_\gamma [1 + \cos(n\gamma - \delta_\gamma)] \\
 & + \sum_{\text{improper}} k_\psi [1 + \cos(n\psi - \delta_\psi)] \\
 & + \sum_{i=1}^{N-1} \sum_{i < j}^N \left\{ 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] \right. \\
 & \left. + \frac{q_i q_j}{4\pi \epsilon_0 r_{ij}} \right\}
 \end{aligned} \tag{4}$$

where the terms have their usual meaning.⁴⁰ The Generalized Amber Force Field (GAFF)⁴¹ was used for [C₄mim]⁺ [Tf₂N][−], with the exception of partial charges and dihedral parameters, which were determined using procedures described previously.⁴² The TraPPE force field⁴³ was used for CO₂ and CH₄, and the Cracknell force field⁴⁴ was used for H₂. For dynamics calculation, CO₂ was rendered flexible using the bond and angle constants from the work of Shi and Maginn.⁴⁵ The carbon atoms in the graphite slit pores were modeled as rigid Lennard-Jones spheres with parameters, $\sigma = 3.40 \text{ \AA}$ and $\epsilon = 0.055 \text{ kcal/mol}$. A complete listing of force field parameters is given in the Supporting Information. Although this type of “Class I” force field has been used previously to model gas absorption and diffusion in ionic liquids, it is important to note that the model lacks some features that could be important in accurately modeling this particular system, including polarizability and flexibility of the graphite walls.

Monte Carlo Simulation

Isothermal-isobaric Gibbs ensemble Monte Carlo (NPT GEMC) simulations were conducted with the code Cassandra^{46,47} to compute the sorption isotherms of CO₂, CH₄, H₂, CO₂/CH₄ and CO₂/H₂ in the graphite slit pores with and without [C₄mim]⁺ [Tf₂N][−] at 333 K and

pressures ranging from 1 bar to 200 bar. Initial configurations of the confined IL systems were generated by placing 250 ion pairs of [C₄mim][Tf₂N] inside the slit pores using a configurational bias methodology.⁴⁶ The widths of the slit pores were determined using molecular dynamics, as described below. During the course of a simulation, thermal equilibration of the system was achieved by displacing the center-of-mass (COM) and rotating about the COM of ions and gas molecules. Biased regrowth moves were also used to sample conformations of the butyl chain of the cations. The pressure in the gas phase simulation box was maintained at the desired value by performing volume displacements, while the volume of the simulation box containing the slit pore was held constant. Chemical potential equilibration for a given gas was established by exchanging gas molecules between the two simulation boxes. The configurational bias Monte Carlo method was employed to increase the efficiency of these exchanges. As the carbon atoms in the graphite slit pores were held fixed, the interaction energy of the ionic liquid and/or gas molecules with the framework was computed using a computationally efficient potential map lookup method.⁴⁸ In this approach, prior to an actual simulation, the simulation box containing the slit pore was discretized into a large number of cubes of volume $8 \times 10^{-3} \text{ \AA}^3$. For each distinct atom type present in the system, the Lennard-Jones potential and its first and second derivatives were computed and tabulated at the each of the grid points of these cubes. The grid points with energies greater than $35 k_B T$ were deemed high energy regions and discarded from tabulation. During a simulation, the Lennard-Jones energy between the graphite and a given atom was calculated from a three-dimensional cubic Hermite interpolation of the energies of the grid points of the cube enclosing the atomic position. For the sorption calculations in empty slit pores, equilibration runs of 20 M steps were performed followed by the production runs of 10 M steps. For confined IL systems, equilibration runs ranging from 200 to 300 M Monte Carlo steps were performed, followed by the production runs of 60 M Monte Carlo steps. The Lennard-Jones and electrostatic interactions for the simulation box containing the slit pore were truncated at $14 \text{ \AA}$. The bond lengths were constrained to their nominal values. Bulk

IL solubilities for these gases were computed in earlier work from our group.⁴⁹

Molecular Dynamics Simulation

The molecular dynamics (MD) package LAMMPS⁵⁰ was used to compute the self-diffusivities of CO₂, CH₄, H₂ and their mixtures in all the systems. For the simulation of confined [C₄mim]⁺ [Tf₂N]⁻, an initial configuration was generated with VMD⁵¹ by inserting 250 cation-anion pairs in a slit pore with x and y dimensions of 5.16 and 5.1 nm, respectively. A gap of 2 Å was kept between the IL and the graphite wall. The system was placed in a periodic orthogonal box with x and y dimensions equal to the graphite sheets. The box length in the z dimension was fixed at 20 nm, long enough to remove any artifacts caused by periodicity. The energy of the system was minimized using a conjugate gradient technique. Next, random velocities were assigned to the ions, and a normal force equivalent to 1 bar pressure along the z axis was applied on one graphite wall while the other wall was fixed, resulting in effectively an isothermal-isobaric ($NP_{zz}T$) ensemble, where P_{zz} is the component of the pressure tensor normal to the pore walls. The temperature of the system was maintained by using the Nosé-Hoover thermostat⁵² with a damping factor of 0.1 ps. No barostat was used because all the MD simulations were performed in the NVT ensemble followed by the NVE ensemble. Pressure was converted to its equivalent force, which was applied to the slit pore wall along the z dimension, while the x and y dimensions of the simulation box were equal to the slitpore dimensions. This allowed the gap between the walls to fluctuate. The system was equilibrated for 10 ns, and the average gap distance between the pore walls was recorded. The last 2 ns of the equilibration run was averaged to obtain the equilibrium pore width of 5.13 nm. Different pressures up to 100 bar were applied to the system. There was no significant change in the pore width as a function of pressure, consistent with the very small compressibility of ILs. The density of IL thus obtained was used as a basis to determine the pore dimensions in the x and y directions for the 2 nm pore. Since the number of ion pairs was kept constant at 250, pore dimensions in x and y

were increased to 8.35 and 8.08 nm for the 2 nm pore. The aforementioned procedure was repeated for a slit pore of smaller width, and an equilibrated width of 2.19 nm was obtained. Note that while we will refer to nominal pore widths of 5 nm and 2 nm, the actual distance between the centers of mass of atoms lining the pores are 5.13 nm and 2.19 nm.

The MC solubility calculations of pure and mixed gases in the empty slit pores and the confined IL systems were used to set the concentrations of gas molecules in the MD simulations. The software package VMD⁵¹ was used to insert 250 IL ion pairs and the corresponding number of gas molecules based on their mole fractions in slit pores of sizes $(5.16 \times 5.1 \times 5.13) \text{ nm}^3$ and $(8.35 \times 8.08 \times 2.19) \text{ nm}^3$. Then the systems were placed in periodic orthogonal boxes with their sizes equal to the slit pore in the x and y dimensions. Similar to the simulation of pure IL in a slit pore, box lengths along the z axis were maintained at 20 nm, long enough to remove any artifacts caused by periodicity. The systems were allowed to equilibrate for 5 ns, using canonical (NVT) ensemble simulations. After equilibration, microcanonical ensemble (NVE) production runs of 60 ns were performed, from which trajectories were taken to compute transport properties.

The Lennard-Jones and real space electrostatic interactions were truncated at 12 Å. Long range electrostatic interactions were handled using the particle-particle- particle-mesh Ewald method.⁵³ Coordinates were recorded every 5 ps for the IL and every 0.1 ps for CO₂, CH₄ and H₂. A 1 fs time-step was used.

Results and Discussion

Density

The local density profiles of IL and CO₂ along the z axis in the confined systems at 333 K and P_{zz} = 10 bar are shown in Figure 1. Both the IL and CO₂ have high density layers adjacent to the pore walls, followed by weaker density oscillations towards the pore center. Similar density profiles were observed for IL-CH₄, and IL-CO₂ mixtures (see Figs. S2 and S3

in Supporting Information). This is typical of confined fluids and has been observed in many simulation studies.^{2,32,33,54} The 5 nm pore is wide enough such that the density fluctuations of the IL decay and a bulk liquid IL density is observed at the center of the pore. The 2 nm pore is too narrow for this to occur, however, and density fluctuations persist across the entire pore.

The CO₂ density profiles exhibit the same type of density fluctuations as the IL in both the slit pores, reaching a constant value at the middle of the 5 nm pore but not the 2 nm pore. Interestingly, the density at the center of the 5 nm pore is greater than the bulk gas CO₂ density and CO₂ density in bulk IL at the same temperature and pressure. The favorable interactions CO₂ experiences with the IL and the pore walls serve to “concentrate” the CO₂ in the pores.

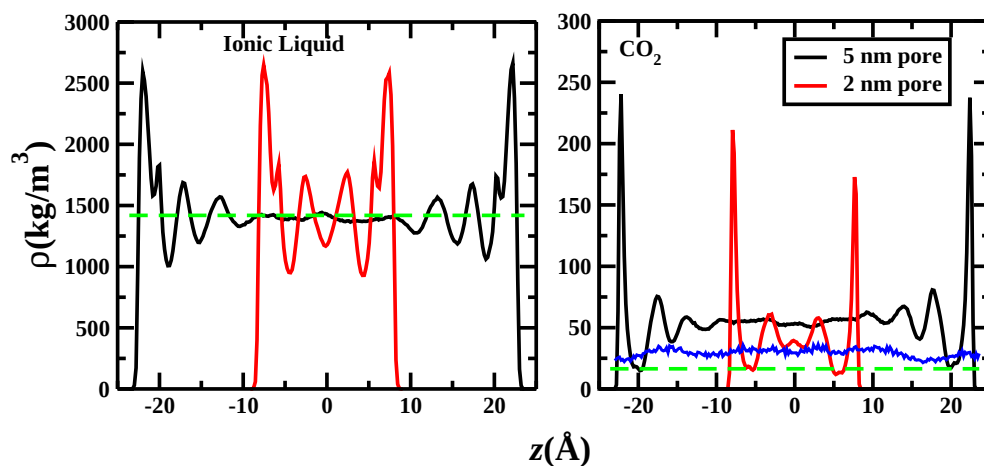


Figure 1: Center of mass (COM) local density profiles of [C₄mim]⁺ [Tf₂N][−] (left) and CO₂ (right) in the confined systems at 333 K and $P_{zz} = 10$ bar. For comparison, bulk densities of pure [C₄mim]⁺ [Tf₂N][−] and CO₂ obtained from NPT simulations are shown as green dashed lines. Local density profile of CO₂ in bulk IL at 10 bar is shown in blue.

We estimated the overall density of IL in the pores with the following equation

$$\rho = \frac{N}{L_x \times L_y \times L_{z_{ef}}} \quad (5)$$

where N is the number of ion pairs, L_x and L_y are lengths of the pores in the x and y dimensions, and $L_{z_{ef}}$ is the effective pore width, taken as the distance between the IL density peaks nearest the pore wall. This definition of volume neglects the small excluded volume right next to the graphite walls. Thus $L_{z_{ef}}$ is 4.65 ± 0.02 nm for the 5 nm pore and 1.78 ± 0.02 nm for the 2 nm pore. Overall IL densities at 333 K and 10 bar were 1449 ± 13 kg/m³ for the 2 nm pore and 1421 ± 6 kg/m³ for the 5 nm pore, which are just slightly higher than the bulk liquid density value of 1419 ± 4 kg/m³. This is consistent with the fact that ILs have very small compressibilities and high cohesive energy densities, and so are minimally compressed upon extreme confinement.

Pure Gas Adsorption

Excess adsorption isotherms (ρ^{excess}) for pure CO₂, CH₄ and H₂ at 333 K in the 5 nm and 2 nm pores (with no IL) are shown in Fig 2. ρ^{excess} is defined as

$$\rho^{\text{excess}} = \rho^{\text{adsorbed}} - \rho^{\text{bulk}} \quad (6)$$

where ρ^{adsorbed} and ρ^{bulk} are gas densities in the pores and in the gas phase, respectively. ρ^{adsorbed} was computed using the pore volume definition given in eq 5. Separate NPT simulations were performed to compute the bulk gas densities of CO₂, CH₄ and H₂.

Figure 2 shows that the 2 nm pore has higher excess adsorption than the 5 nm pore for all gases. This is because a greater fraction of the gas molecules interact with the pore walls in the 2 nm pore. This is consistent with several other studies on the adsorption of CO₂,^{55,56} CH₄^{57,58} and H₂^{59,60} in graphite slit pores. For both pores, excess adsorption follows the order CO₂ > CH₄ > H₂, consistent with the relative interaction energy between the graphite and the gases. The CO₂ isotherms are linear up to 10 bar. Due to their weaker interactions with the pore walls, much higher pressures were simulated for CH₄ and H₂. At the very highest pressures, the CH₄ and H₂ isotherms show curvature, indicating the beginning of

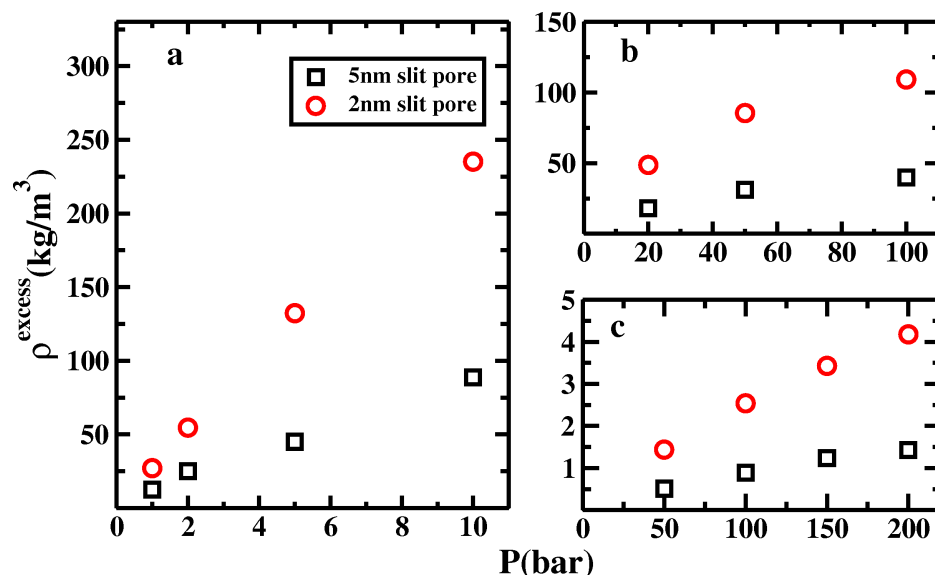


Figure 2: Excess adsorption of CO₂ (a), CH₄ (b) and H₂ (c) in 5 nm and 2 nm slit pores at 333 K and different pressures.

saturation.

Calculated CO₂, CH₄ and H₂ solubilities in the bulk IL and the confined IL systems are shown in Figure 3. Numerical values are listed in Table S9 in the Supporting Information. Mole fraction is defined as the moles of gas absorbed divided by the sum of the moles of IL and moles of gas. The gas solubilities are larger in the confined IL systems than in bulk IL. CO₂ solubilities up to 5 bar, CH₄ solubilities at 20 bar, and H₂ solubilities at 50 bar are essentially the same in both pores, unlike what was observed for the empty pores. At higher pressures, however, gas solubilities are slightly larger in the 5 nm pore. CO₂ has significantly higher solubility at a given pressure than either CH₄ or H₂. Figure 4 shows snapshots of CO₂ absorbed in the confined 5 nm and 2 nm systems. Consistent with the density profiles in Fig. 1, CO₂ localizes near the pore wall, but is also distributed throughout the liquid. The favorable interactions with both the pore wall and the IL lead to higher overall densities of CO₂ in the confined systems than in bulk IL.

Simulation studies performed by Shi et al.^{31,32} on the solubility of CO₂ and H₂ in IL confined in carbon nanotubes and silica nanopores have shown that confinement enhances CO₂ and H₂ solubility. Similarly, Zhang et al.³⁶ through experimental investigation of CO₂

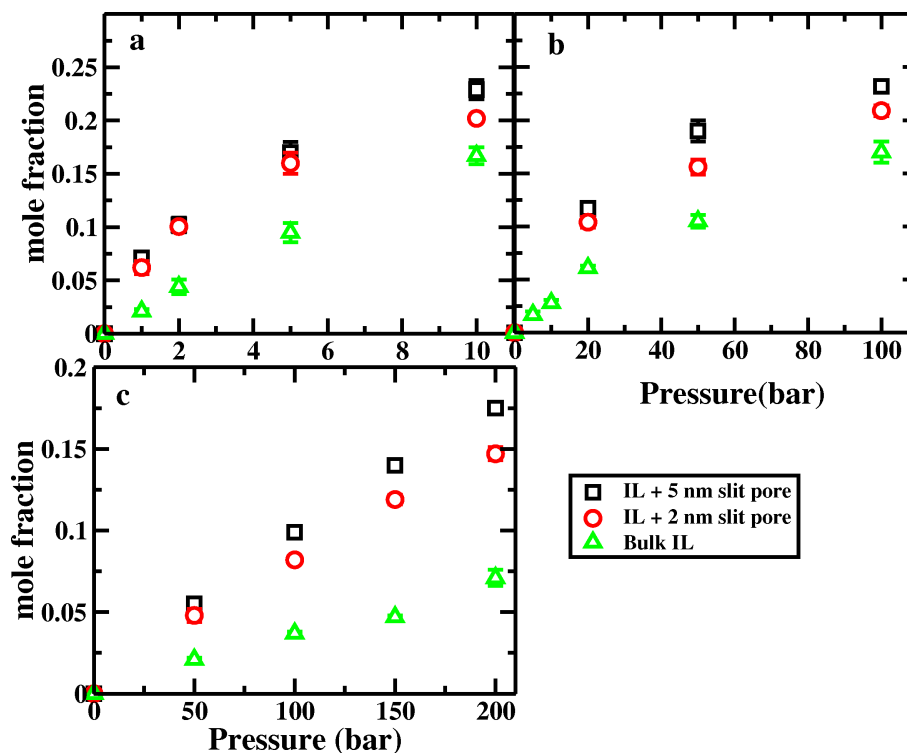


Figure 3: Solubility of CO_2 (a), CH_4 (b) and H_2 (c) in bulk $[\text{C}_4\text{mim}]^+ [\text{Tf}_2\text{N}]^-$ and $[\text{C}_4\text{mim}]^+ [\text{Tf}_2\text{N}]^-$ confined in 5 nm and 2 nm slit pores at 333 K and different pressures. CO_2 and CH_4 solubility in bulk $[\text{C}_4\text{mim}]^+ [\text{Tf}_2\text{N}]^-$ are from our previous work.⁶¹ H_2 solubility results in bulk $[\text{C}_4\text{mim}]^+ [\text{Tf}_2\text{N}]^-$ are from the work of Singh et al.⁴⁹

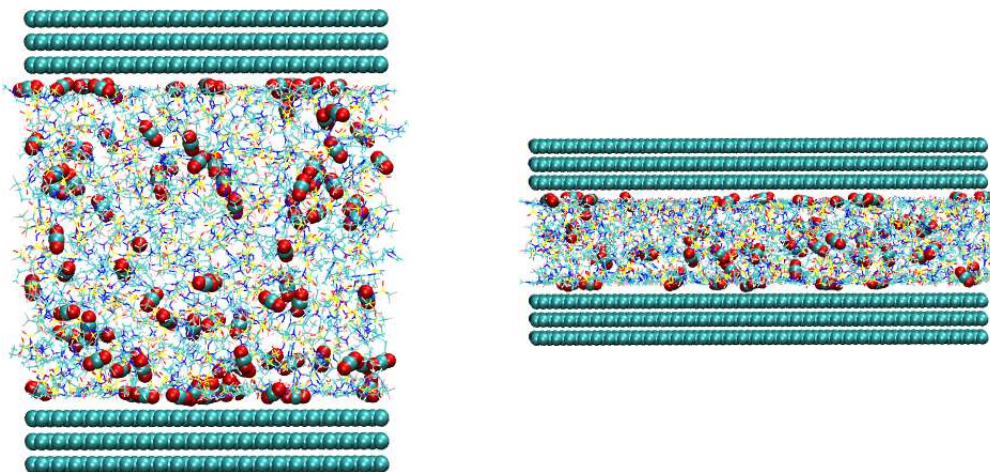


Figure 4: Snapshots $[\text{C}_4\text{mim}]^+ [\text{Tf}_2\text{N}]^-$ - CO_2 mixtures confined in 5 nm graphite slit pore (left) and 2 nm graphite slit pore (right) at 10 bar and 333 K.

solubilities in $[\text{C}_4\text{mim}]^+ [\text{BF}_4]^-$ confined in silica gel pores of 3-12 nm, have demonstrated that CO_2 solubility increases by a factor of 1.5 relative to the bulk IL. Both of these results

are consistent with our results. While our results show that at high pressure CO₂ and H₂ solubilities are higher in the 5 nm pore than in the 2 nm pore, Shi and Luebke³² observed the opposite trend for CO₂ and H₂ solubilities in [C₆mim]⁺[Tf₂N]⁻ confined in silica slit pores. They observed that the solubility was higher in the smaller pores. The reason for this increase in solubility with decreasing pore width was attributed to the fact that the IL density decreased with decreasing pore width, thereby providing more free volume. This is in contrast to our system where IL density increased with decreasing pore width. The differences between the trends observed here and those observed by Shi and Luebke may be due to the differences in the way confined systems were prepared or to differences in the interaction energies of silica and graphite. It is also possible that some of the differences are simply due to density oscillations induced by the walls and whether or not the pore width is commensurate with a maximum or minimum density peak. At certain pore widths the ions pack well and at others they don't. This makes comparison with simulations performed on other systems difficult. What is consistent in both cases is that confined systems with reduced IL density will absorb more gas.

Interaction Energies

Figure 5 shows the interaction energies (electrostatic + van der Waals) that CO₂ (top, 10 bar and 333 K) and CH₄ (bottom, 50 bar and 333 K) experience in the confined IL systems. IL-gas interactions dominate, accounting for about 80 % of the total interaction energy in both pores. This is consistent with the fact that the density profiles of the gases track the density profiles of the IL, showing areas of crests and troughs at the same distance from the walls. Cation-gas and anion-gas interaction energies are similar for both pores, while gas-gas interaction energies in both pores are negligible. Wall-gas interactions are about twice as large in the 2 nm pore as they are in 5 nm pore, but the surface area of the 2 nm pore is also about twice as large as the surface area of the 5 nm pore. When the gas-wall energies are normalized by surface area, the energies are nearly the same for the two pores.

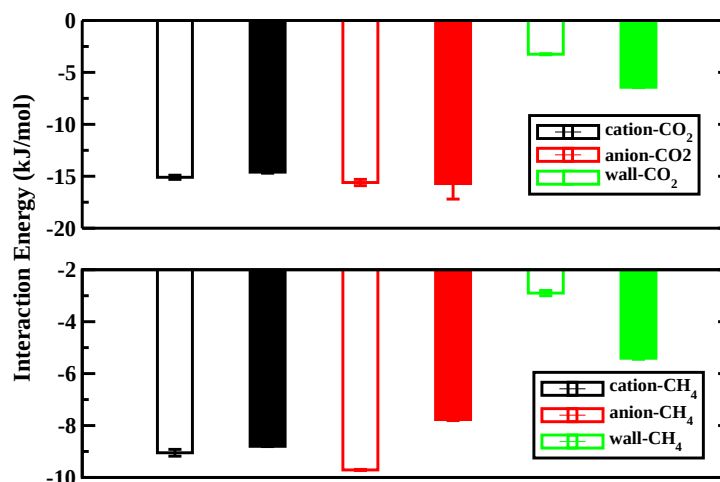


Figure 5: CO_2 - $[\text{C}_4\text{mim}]^+ [\text{Tf}_2\text{N}]^-$ (top) and CH_4 - $[\text{C}_4\text{mim}]^+ [\text{Tf}_2\text{N}]^-$ (bottom) interaction energies confined in 5 nm pores (open bars) and 2 nm pores (solid bars) at 333 K. The pressure of the CO_2 system is 10 bar and is 50 bar for the CH_4 system. Gas-gas interaction energies are not shown because they are negligible.

For CO_2 the energies are $-0.095 \text{ kJ/mol nm}^2$ and $-0.123 \text{ kJ/mol nm}^2$ in the 2 nm and 5 nm pores, respectively. For CH_4 the energies are $-0.081 \text{ kJ/mol nm}^2$ and $-0.11 \text{ kJ/mol nm}^2$ in the 2 nm and 5 nm pores, respectively. For an empty pore, the higher surface area of the small pore leads to increased adsorption relative to the large pore. When IL fills the pore, however, greater IL density in the 2 nm pore offsets this, and both pores have similar isotherms until the very highest pressures. We find that IL-wall interactions decrease as pore width increases, from $-53.4 \pm 0.1 \text{ kJ/mol}$ in the 2 nm pore to $-19.4 \pm 0.2 \text{ kJ/mol}$ in the 5 nm pore. This is consistent with the differences in surface area between the two pores. A similar trend was observed by Shi and Luebke³² for silica slit pores.

Pure Gas Self-Diffusivity

Self-diffusion coefficients for gas and gas-IL mixtures under confinement were computed from the Einstein relation

$$D = \lim_{t \rightarrow \infty} \frac{1}{2dt} \langle |r(t) - r(0)|^2 \rangle \quad (7)$$

where $r(t)$ is the position of a species at time t and $r(0)$ is the position at time $t = 0$. The term in brackets $\langle \dots \rangle$ is the mean squared displacement (MSD) and d is the dimension of the system. Since the displacement in a slit pore is limited in the z direction, a value of $d = 2$ was used to compute the self-diffusion coefficient in the x and y directions. Figures 6 and 7 show the results.

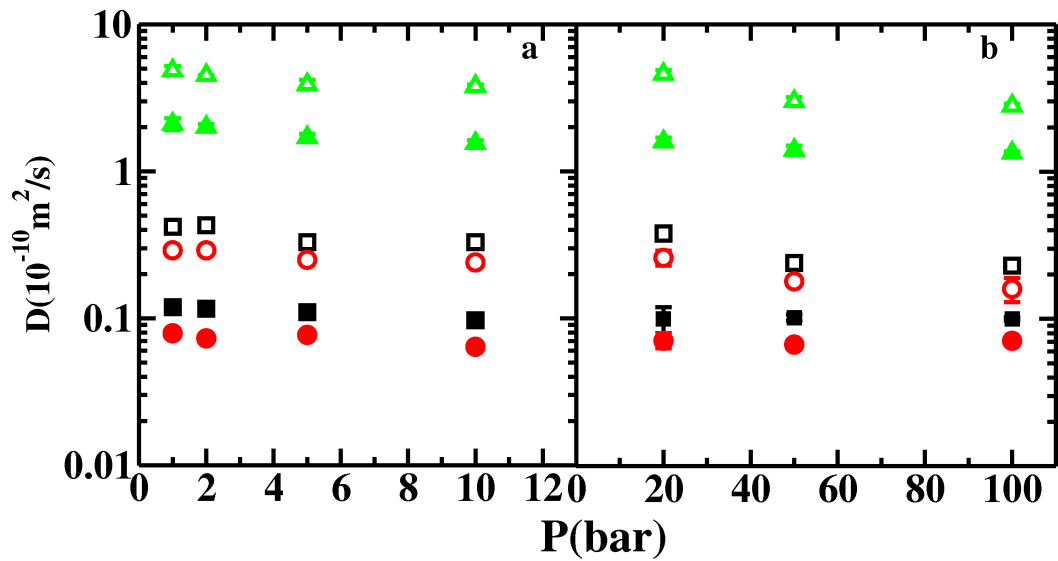


Figure 6: CO_2 - $[\text{C}_4\text{mim}]^+ [\text{Tf}_2\text{N}]^-$ (a) and CH_4 - $[\text{C}_4\text{mim}]^+ [\text{Tf}_2\text{N}]^-$ (b) self-diffusion coefficients in 5 nm (open symbols) and 2 nm (closed symbols) slit pores at 333 K and different pressures. Green symbols represent the gas, black symbols the cation, and red symbols the anion.

The self-diffusion coefficients of the cation, anion and gases are greater in the 5 nm pore by a factor of about 3-4 compared to the 2 nm pore. Also, the self-diffusion coefficients of gases are larger than that of the cation and anion. Typical of this class of ILs, the self-diffusion coefficients of the cation are greater than that of anion.^{62,63} For the pressures studied, the self-diffusion coefficients of the cations, anions, CO_2 and CH_4 decrease slightly with increasing pressure. In a dynamics study of $[\text{C}_4\text{mim}]^+ [\text{PF}_6]^-$ confined in a 5 nm graphite slit pore, Singh et al.³³ have shown that the IL in the 5 nm graphite slit pore has a bulk-like region where the self-diffusion coefficients of the ions are similar to the bulk phase, but have reduced self-diffusion coefficients near the pore walls. Similar heterogeneity in the dynamics of IL was observed in the present system (not shown). The self-diffusion

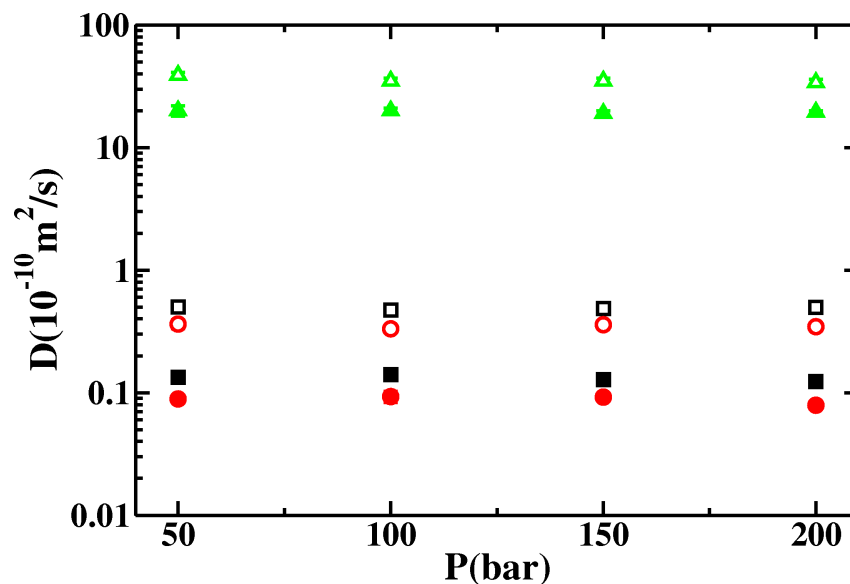


Figure 7: H_2 - $[\text{C}_4\text{mim}]^+[\text{Tf}_2\text{N}]^-$ self-diffusion coefficients in 5 nm (open symbols) and 2 nm (closed symbols) slit pores at 333 K and different pressures. Green symbols represent the gas, black symbols the cation, and red symbols the anion.

coefficient of IL in the 5 nm pore is comparable to that of IL in the bulk phase (see Table S10 in Supporting Information). For IL in the 2 nm pore, however, there is no bulk-like region. Because of the proximity of the walls, ILs and gases experience greater energy of interaction from the pore walls, *vide supra*. The IL in the 2 nm pore also has a greater density compared to the IL in the 5 nm pore, which reduces the IL self-diffusion coefficient. Computational studies on dynamics of IL-gas mixtures in carbon nanotubes and silica slit pores by Shi and Sorescu³¹ and Shi and Luebke³² have shown enhanced self-diffusion coefficients of $[\text{C}_6\text{mim}]^+[\text{Tf}_2\text{N}]^-$ and gases such as CO_2 , H_2 and N_2 . They observed a 25-30 % reduction in IL-IL pair interaction energies in carbon nanotubes compared to that in the bulk phase, which they attributed to be the cause of increased self-diffusivities. In our case, IL-IL pair interaction energies were computed to be -188 ± 1 , -175 ± 1 and -161 ± 1 kJ/mol per IL pair in the bulk, 5 nm and 2 nm pores, respectively at 20 bar and 333 K. Hence, confinement led to the reduction of IL-IL interaction energies by only 13 % and 6 % in the 2 nm and 5 nm pores when compared to the bulk phase. Since the self-diffusion coefficients of the IL in the 5 nm pore and bulk phase are identical while they are smaller in 2 nm pores, it suggests

that IL-graphite interaction energies play a role in determining the self-diffusion coefficients only for the very smallest pores. In the case of silica nanopores, low IL density and the higher self-diffusion coefficients of the IL than the bulk phase were observed at temperatures below 373 K. Similarly, an experimental study by Iacob et al.³⁷ has shown an increase in dynamics of $[\text{C}_4\text{mim}]^+ [\text{BF}_4]^-$ by two orders of magnitude when confined in silica nanopores. Such an increase in the ionic mobility was attributed to the changes in ionic packing due to reduction of the IL density. However, we observe that the IL confinement in a graphite slit pore increases its density which reduces the self-diffusion coefficient. These results suggest that the dynamics of ILs are a function of the type of confining material, and that no simple rules can be used to predict the behavior of these systems.

The self-diffusion coefficients of CO_2 , CH_4 and H_2 in the bulk (gas) phase and slit pores without IL are shown in Figure 8. Gas diffusivities in the empty pores are about four orders of magnitude greater than they are in the pores containing IL. Increased density of the gases under confinement leads to their decreased self-diffusivities when compared to the bulk gas phase. In addition, wall-gas interactions further reduce the dynamics of the gases. CO_2 self-diffusion coefficients are more sensitive to confinement than CH_4 and H_2 , due to its stronger interaction with the pore walls.

Mixtures of CO_2/CH_4 and CO_2/H_2

To determine the permselectivity of CO_2 over CH_4 and H_2 , solubility selectivities and diffusion selectivities were computed for gas mixtures under different conditions. NPT GEMC simulations of CO_2/CH_4 and CO_2/H_2 mixtures at a temperature of 333 K and total pressures of 100 bar and 30 bar, respectively, were carried out to compute solubility selectivities. The initial $\text{CO}_2:\text{CH}_4$ gas phase mole ratio was 5:95 while for $\text{CO}_2:\text{H}_2$ it was 15:85. Note that these values fluctuate as the amount of adsorbed / absorbed gas is different for each case. Diffusion selectivities were computed by running MD simulations of gas mixtures at con-

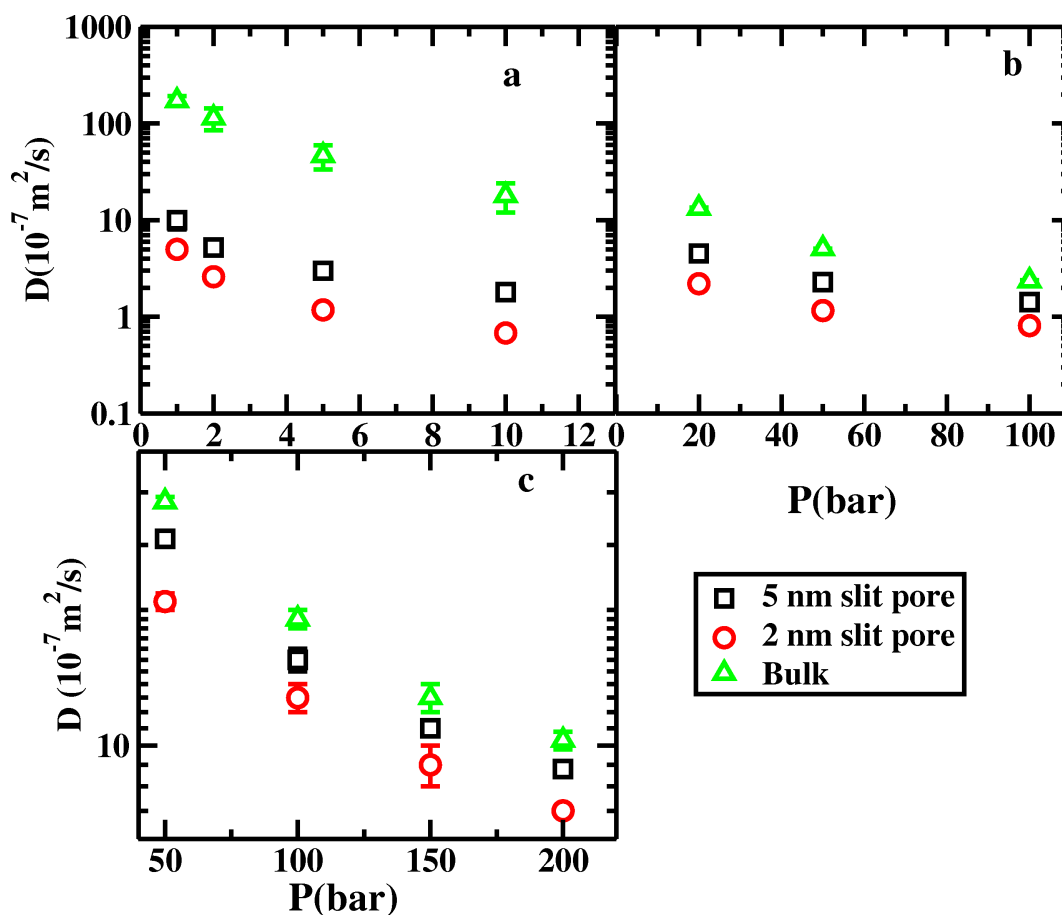


Figure 8: CO_2 (a), CH_4 (b) and H_2 (c) self-diffusion coefficients in bulk, 5 nm, and 2 nm empty slit pores at 333 K and different pressures.

centrations obtained from the GEMC simulations. Table 1 shows the computed equilibrium gas phase and sorbed phase mole fractions for each species in the five different systems considered along with the calculated self-diffusivities. Table 2 shows the solubility selectivities, diffusion selectivities and permselectivities for each system.

While there are small differences in the total amount of CO_2 , CH_4 and H_2 absorbed (or adsorbed) in the different systems, there are two striking features of the solubility results. First, there is a much larger solubility selectivity of CO_2 for the systems containing IL relative to the empty pore system. This is due to the stronger association between the IL and CO_2 relative to the other gases, as shown in Fig. 5. Whether in the bulk liquid phase or confined in nanopores, the IL serves to improve the solubility selectivity of CO_2 over CH_4 and H_2 .

Second, confinement slightly increases the solubility selectivity relative to the bulk IL, but the effect is small. The 2 nm empty pore has a higher solubility selectivity than the 5 nm pore for CO₂ over H₂, but in the CO₂/CH₄ system, the two pore sizes have essentially the same solubility selectivity.

Diffusion selectivities were computed from mixed gas MD simulations, and the results are shown in Table 2. The self-diffusivities of CO₂ and CH₄ are essentially the same in the bulk IL and in the pores containing IL, such that the diffusion selectivity is close to unity. In the empty pores, CH₄ diffuses faster than CO₂ so that the diffusion selectivity is less than unity. The result is that for the systems containing IL, the permselectivities are mostly determined by the solubility selectivities, while for the empty pores the modest solubility selectivity of CO₂ over CH₄ is nearly offset by the poor diffusion selectivity; permselectivities are not much greater than unity in the 2 nm pore and are even less than unity in the 5 nm pore.

In the case of CO₂ and H₂, the high solubility selectivity for CO₂ in the systems containing IL is somewhat offset by the diffusion selectivities of 0.10-0.18, which favors permeation of H₂ over CO₂. The resulting permselectivities are around 10, similar to those of CO₂ over CH₄. The empty pores have similar diffusion selectivities as the systems containing IL but their lower solubility selectivity means that they are much less effective than the IL systems at separating CO₂ from either CH₄ or H₂.

The self-diffusivities of the gases in the empty pores are much larger than for the systems containing IL, and confinement of the IL reduces the self-diffusivities even more relative to the bulk IL. Because each gas is slowed by about the same amount, however, confinement and the presence of the IL does not have a major effect on diffusion selectivity. If the rate limiting step for permeation is diffusion within the pores, then the current results suggest that the use of confined ILs will lower overall permeation rates relative to empty porous materials. If, on the other hand, the gas desorption process is rate limiting, then the use of confined ILs will lead to greatly improved permselectivities with little effect on overall

permeability.

Table 1: CO₂/CH₄ and CO₂/H₂ solubilities and self-diffusion coefficients ($D \times 10^{10} \text{ m}^2/\text{s}$) in different systems at 333 K and a total pressure of 100 bar (top) and 30 bar (bottom). x is the mole fraction in adsorbed or absorbed phase and y is the mole fraction in gas phase. Subscripts are the uncertainty in the last digit.

System	y_{CO_2}	y_{CH_4}	x_{CO_2}	x_{CH_4}	D_{cation}	D_{anion}	D_{CO_2}	D_{CH_4}
IL + 2 nm pore	0.046 ₀	0.954 ₁	0.061 ₈	0.167 ₅	0.087 ₂	0.059 ₃	1.4 ₁	1.25 ₆
IL + 5 nm pore	0.037 ₀	0.963 ₄	0.069 ₉	0.208 ₁	0.32 ₁	0.221 ₉	4.3 ₄	3.8 ₁
IL*	0.038 ₁	0.962 ₂	0.044 ₃	0.179 ₇	0.57 ₆	0.42 ₄	6.4 ₆	6.1 ₅
2 nm pore	0.043 ₀	0.956 ₀	0.079 ₁	0.921 ₁	—	—	453 ₁₀	520 ₅₇
5 nm pore	0.047 ₀	0.953 ₀	0.071 ₀	0.928 ₁	—	—	610 ₂₁	798 ₁₅

System	y_{CO_2}	y_{H_2}	x_{CO_2}	x_{H_2}	D_{cation}	D_{anion}	D_{CO_2}	D_{H_2}
IL + 2 nm pore	0.09 ₁	0.91 ₁	0.116 ₈	0.015 ₁	0.12 ₇	0.07 ₁	1.9 ₁	18 ₁
IL+ 5 nm pore	0.088 ₂	0.91 ₀	0.118 ₃	0.02 ₀	0.41 ₄	0.29 ₆	4.3 ₃	32 ₃
IL**	0.138 ₁	0.862 ₁	0.086 ₆	0.009 ₁	0.50 ₆	0.37 ₃	5.5 ₂	46 ₅
2 nm pore	0.097 ₀	0.953 ₀	0.455 ₂	0.545 ₂	—	—	1852 ₁₂₃	10230 ₁₈₁₂
5 nm pore	0.043 ₀	0.956 ₀	0.269 ₁	0.730 ₀	—	—	2870 ₃₃₅	19228 ₁₃₉₃

*Results are from our previous work.⁶¹ ** Solubility results are from the work of Singh et al.⁴⁹

Note that single component (pure gas) solubilities and diffusivities are often used to estimate mixture selectivities, because mixture properties are difficult to obtain. That is, pure gas values of D_i and S_i are used to estimate P_i in eq 1. This assumes that the system behaves ideally such that the gas molecules sorb and diffuse independent of the presence of the other species. We find that for the confined IL systems, the ideal assumption gives a reasonably accurate estimate of the diffusion selectivity for these systems but is not reliable for estimating solubility selectivities of CO₂ over CH₄ and H₂. For example, using pure gas diffusion and solubility data (Table S9 of the Supporting Information) diffusion selectivities for CO₂ over CH₄ were 1.2 and 1.3 for the 2 nm and 5 nm pores, respectively. This is statistically equivalent to the value of 1.2 obtained for both pores from mixture simulations. The same is true for the diffusion selectivity of CO₂ over H₂; pure gas values range from 0.09 to 0.11, in agreement with the mixed gas values of 0.107 and 0.14. On the other hand,

Table 2: CO₂/CH₄ and CO₂/H₂ permselectivities in empty slit pores, bulk IL and confined IL systems at 333 K and total pressures of 100 and 30 bar, respectively. Also shown are solubility selectivities and diffusion selectivities. Subscripts are the uncertainty in the last digit.

Pressure	System	y_{CO_2}	y_{CH_4}	β_{CO_2/CH_4}^S	β_{CO_2/CH_4}^D	β_{CO_2/CH_4}^P
100	IL + 2 nm pore	0.046 ₁	0.954 ₁	8 ₁	1.2 ₁	9 ₂
100	IL + 5 nm pore	0.037 ₁	0.963 ₃	8.7 ₄	1.2 ₁	10 ₁
100	IL*	0.038 ₁	0.962 ₂	6.2 ₈	1.05 ₅	6.4 ₉
100	2nm pore	0.043 ₀	0.956 ₀	1.89 ₅	0.87 ₃	1.6 ₁
100	5 nm pore	0.047 ₀	0.953 ₀	1.57 ₂	0.76 ₄	1.2 ₁

Pressure	System	y_{CO_2}	y_{H_2}	β_{CO_2/H_2}^S	β_{CO_2/H_2}^D	β_{CO_2/H_2}^P
30	IL + 2 nm pore	0.09 ₁	0.91 ₁	80 ₁₂	0.107 ₅	9 ₂
30	IL + 5 nm pore	0.088 ₂	0.91 ₀	61 ₂	0.14 ₂	9 ₂
30	IL**	0.138 ₁	0.862 ₁	59 ₂	0.18 ₂	11 ₂
30	2 nm pore	0.097 ₀	0.953 ₀	8.2 ₁	0.18 ₁	1.5 ₁
30	5 nm pore	0.043 ₀	0.956 ₀	4.1 ₁	0.148 ₇	0.61 ₂

*Results are from our previous work.⁶¹ ** Solubility and Solubility selectivity results are from the work of Singh et al.⁴⁹

ideal solubility selectivities do not agree with their mixture counterparts. Ideal solubility selectivities for CO₂ over CH₄ at the conditions in Table 2 are around 14, significantly higher than the mixture solubility selectivities of 8. Ideal solubility selectivities for CO₂ over H₂ are on the order of 30, which is significantly lower than the mixture values that range from 60-80. This suggests that in these mixtures, there is significant non-ideality in the mixed gas absorption and that great caution should be used when estimating solubility selectivities in ionic liquids using only pure gas solubility data.

Summary and Conclusion

MD and MC simulations were used to compute the self-diffusion coefficients and solubilities of CO₂, CH₄, H₂ and their mixtures in graphite slit pores of 2 nm and 5 nm widths at 333 K and varying pressures. The simulations were repeated for a system in which the ionic

liquid 1-n-butyl-3-methylimidazolium bis(trifluoromethyl sulfonyl)imide ($[\text{C}_4\text{mim}]^+[\text{Tf}_2\text{N}]^-$) was confined within the pores. The results were compared with those obtained for the corresponding gases in the bulk IL phase.

Gas sorption in empty slit pores was enhanced upon decreasing the pore width and is due to the greater gas-surface interactions in the small pores. Gas adsorption followed the order $\text{CO}_2 > \text{CH}_4 > \text{H}_2$ owing to the stronger graphite- CO_2 interaction than that for graphite- CH_4 and graphite- H_2 .

For single component gases, solubilities of CO_2 , CH_4 and H_2 were larger in the confined IL systems than the corresponding bulk IL phase, consistent with the results from previous studies. CO_2 and CH_4 solubilities in the confined IL systems showed little dependence on pore size at lower pressures, while at higher pressures more CO_2 and CH_4 absorbed in the 5 nm pore than in the 2 nm pore. H_2 solubilities were larger for the 5 nm confined IL system than the 2 nm system at all pressures. Lower IL density and the presence of a bulk-like region in the 5 nm pore were found to be responsible for the greater gas solubilities in the 5 nm pore. For confined IL systems, IL-gas interaction energy accounted for about 80% of the total interaction energy experienced by the gases.

The self-diffusion coefficients of gases in the 2 nm pore containing IL were reduced by about 3-4 times when compared to diffusion in the bulk IL and 5 nm pore containing IL. Greater IL-surface and gas-surface interactions in the smaller pore and greater liquid densities were responsible for this. As expected, the self-diffusivities of the gases in the empty pores were three to four orders of magnitude higher than for the cases when IL was present.

Gas sorption and self-diffusivities were also computed for CO_2/CH_4 and CO_2/H_2 mixtures. The mixed gas self-diffusivities followed trends similar to that observed for pure gas self-diffusivities in all systems. Diffusion selectivities were close to unity for CO_2/CH_4 and were around 0.1 for CO_2/H_2 . The presence of the IL significantly increased solubility selectivity for CO_2 over both CH_4 and H_2 . Confinement of the IL led to slightly higher solubility selectivities than that observed in the bulk IL.

The use of pure gas absorption and diffusion results can lead to significant errors in the estimation of the permselectivity of CO₂ over CH₄ and H₂. This is due to the fact that the mixed gas absorption exhibits non-ideality, which results in the ideal assumption overestimating CO₂ absorption selectivity over CH₄ but underestimating CO₂ selectivity over H₂. Ideal diffusion selectivities, on the other hand, agree well with those obtained from mixture diffusivities.

This work has shown that by combining molecular dynamics with Monte Carlo calculations, pure and mixed gas permselectivities can be computed for realistic models. In the bulk liquid phase or when confined in pores on a micron scale or larger, the ionic liquid 1-n-butyl-3-methylimidazolium bis(trifluoromethyl sulfonyl)imide is predicted to yield permselectivities of about 6 for CO₂ over CH₄ and 9 for CO₂ over H₂. By confining the ionic liquid in pores on the nanometer length scale, the permselectivity of CO₂ over CH₄ is expected to increase modestly, while there is little difference for CO₂ over H₂.

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Associated Content

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website.

Ionic liquid structures, atom labels, force field parameters, additional density profiles, excess densities, solubilities and self-diffusivities.

Author Information

Corresponding Author

*E-mail: ed@nd.edu. Tel: (+01) 574-631-5687.

ORCID

Edward J. Maginn: 0000-0002-6309-1347

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